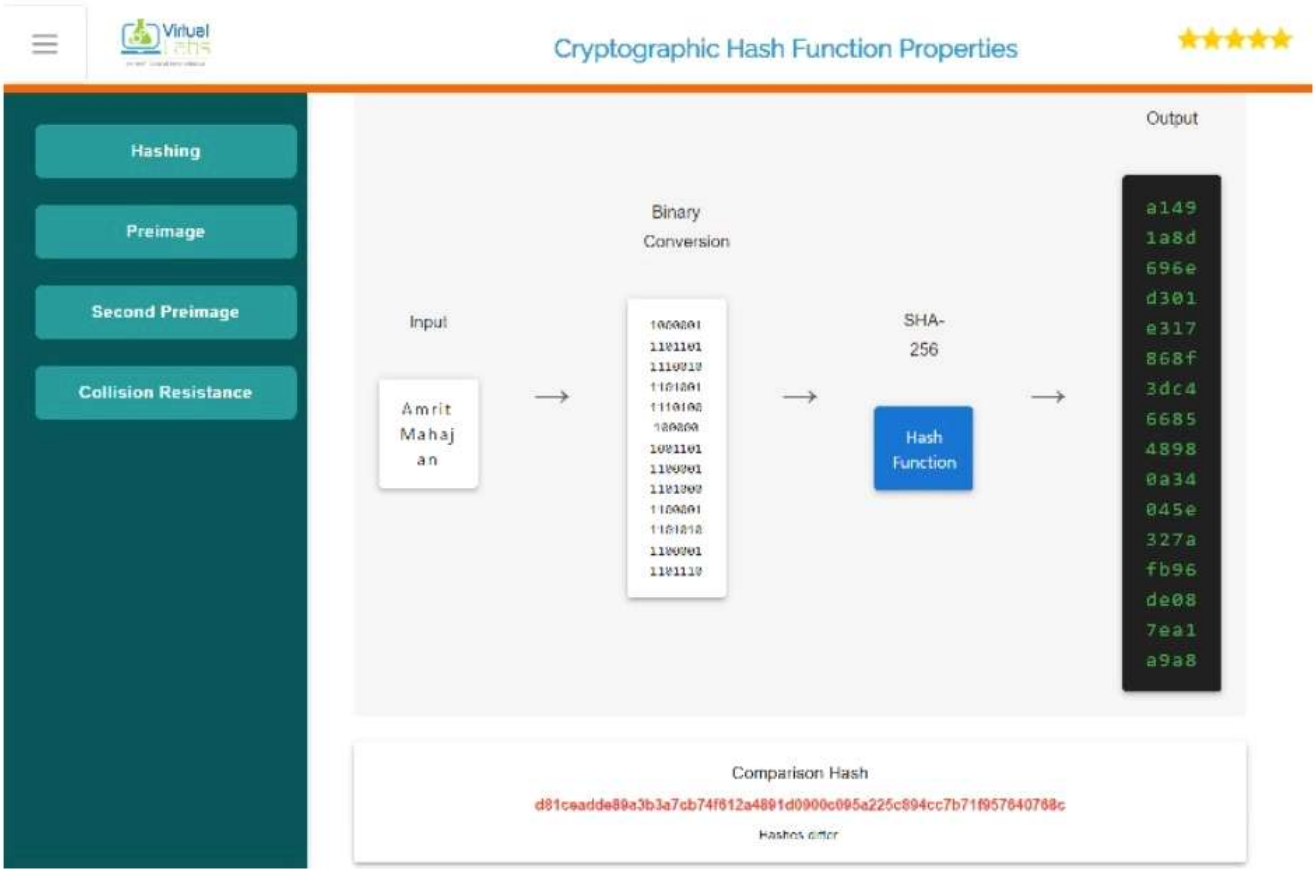
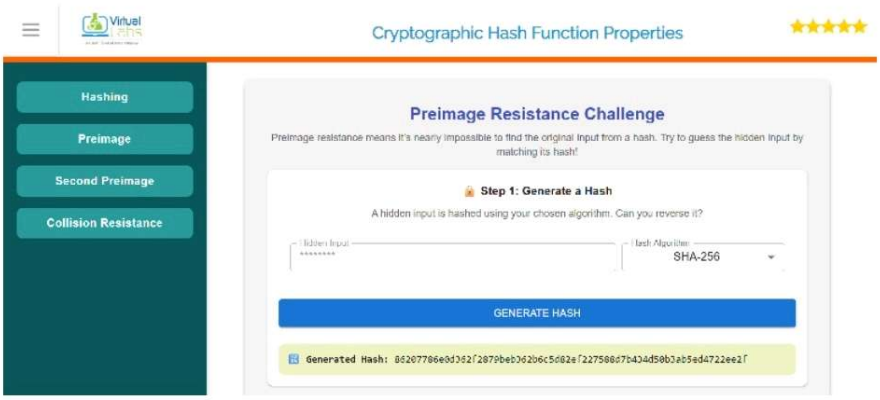
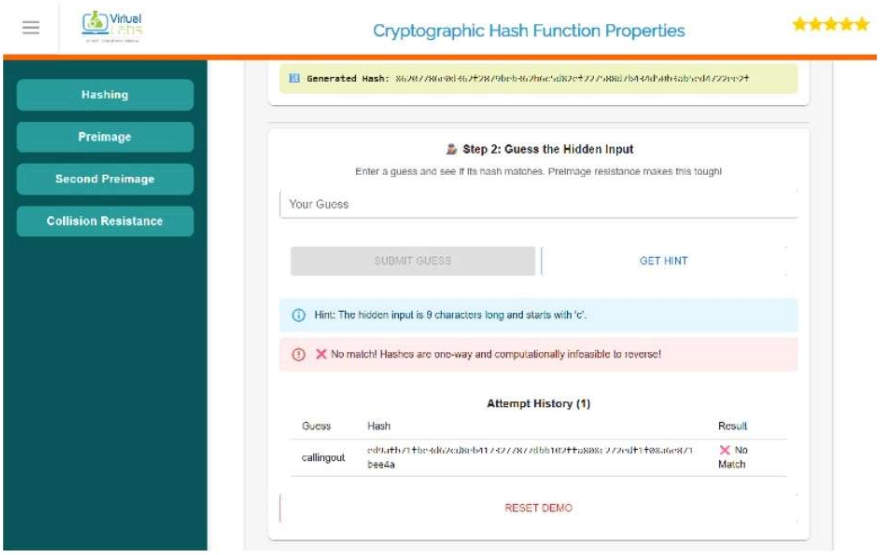
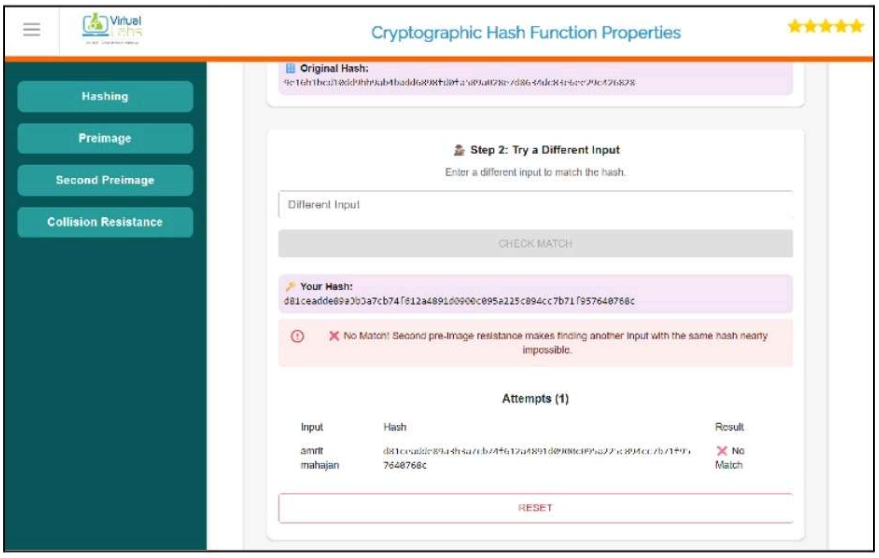

Virtual Labs
Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Cryptographic Hashing Experiment
Explore the properties of cryptographic hash functions by entering text and comparing hashes.
Enter Text
Amrit Mahajan
RANDOM INPUT
Compare Text
amrit mahajan
Hashing Algorithm
SHA-256


Virtual Labs
Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Output
Binary Conversion
Input
Amrit Mahaj an
SHA-256
Hash Function
a149
1a8d
696e
d301
e317
868f
3dc4
6685
4898
0a34
045e
327a
fb96
de08
7ea1
a9a8
Comparison Hash
d81ceadde89a3b3a7cb74f812a4891d0900c095a225c894cc7b71f957640768c
Hashes differ


Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Preimage Resistance Challenge
Preimage resistance means it's nearly impossible to find the original input from a hash. Try to guess the hidden input by matching its hash!
Step 1: Generate a Hash
A hidden input is hashed using your chosen algorithm. Can you reverse it?
Hidden Input
Hash Algorithm
SHA-256
GENERATE HASH
Generated Hash: 86207706e0d362f2079beb362b6c5c02ef227508d7b434d50b3ab5ed4722ee2f

Page Break


Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Step 2: Guess the Hidden Input
Enter a guess and see if its hash matches. Preimage resistance makes this tough!
Your Guess
SUBMIT GUESS
GET HINT
Hint: The hidden input is 9 characters long and starts with 'c'.
No match! Hashes are one-way and computationally infeasible to reverse!
Attempt History (1)
Guess
Hash
Result
callingout
No Match
RESET DEMO


Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Original Hash:
Step 2: Try a Different Input
Enter a different input to match the hash.
Different Input
CHECK MATCH
Your Hash:
d81ceadde89a3b3a7cb74f812a489160900c095a225c894cc7b71f957640768c
No Match! Second pre-image resistance makes finding another input with the same hash nearly impossible.
Attempts (1)
Input
Hash
Result
amrit mahajan
No Match
RESET


Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Step 1: Original Input & Hash
Match this hash with a different input.
Original Input
secure_message_123
Hash Algorithm
SHA-256
Original Hash: